

Assignment-3

Trends and Cloud Services Overview

1. Benefits of serverless Architecture

Serverless architecture allows developers to build and run applications without managing infrastructure.

- Scalability: Automatically scales with demand.
- Cost efficiency: Pay only for actual usage, reducing cost.
- Reduced Operational Overhead: No need to manage server, allowing developers to focus on code.
- Faster Time-to-Market: Accelerates development and deployment cycles.

2. Progressive web apps (PWAs)

PWAs are web applications that offer native app-like experiences.

Key Feature:

- Offline Access: Functionalities without an internet connection.
- Installability: can be added to the user's homescreen.
- Responsive design: Adapt to various screen size and device.
- Push Notification: Engages user with timely updates.

3. Role of AI and Machine Learning in software Architect.

AI and ML are transforming software Architecture by:

- Enhancing User Experience: Personalized recommendation and intelligent interface.
- Automating Process: Predictive analytic and decision-making support.
- Improving Security: Anomaly detection and threat prediction.
- Optimizing Performance: Resources allocation and system tuning based on usage patterns.

4. Cloud computing Service Models

- Infrastructure as a Service [IaaS]: Provided virtualised computing resources over the Internet.
 - Use case: Flexible and scalable hosting solution.
- Platform as a service [PaaS]: Offers to platform allowing customer to develop, run and manage application.
 - Use case: streamlined application development without managing underlying infrastructure.
- Software as a Service [SaaS]: Delivers software application over the internet on subscription basis.
 - Use case: Accessible and maintenance free software solⁿ for end users.